

Introduction

Metallic and oxide nanoparticles play a central role in catalysis, energy storage, biomedicine, and advanced materials, where functional properties depend sensitively on particle size, shape, and polydispersity. Quantitative characterization of these morphological parameters is essential for process optimization and for establishing reproducible structure–property relationships. Among available techniques, transmission electron microscopy (TEM) and scanning TEM (STEM) remain the most widely used methods for direct imaging at sub-nanometer resolution, providing the spatial detail necessary for single-particle measurement [refs].

Despite the maturity of TEM as an imaging modality, the downstream analysis step of extracting quantitative morphological data from micrographs, remains largely manual. The standard workflow in most laboratories relies on general-purpose tools such as ImageJ/Fiji [ref], where researchers trace particle boundaries by hand and record measurements one particle at a time. For the statistically meaningful sample sizes of hundreds to thousands of particles typically needed, this process can consume hours per micrograph. In practice, the annotation cost constrains the number of particles characterized: many studies report distributions over smaller populations than would be desirable simply because the manual effort is prohibitive. This bottleneck is not a limitation of the imaging instrument but of the analysis software available to researchers.

Deep learning has substantially improved automated segmentation of microscopy images. Convolutional neural networks and, more recently, prompt-able foundation models such as the Segment Anything Model (SAM) [ref] have demonstrated strong performance on particle detection and instance segmentation tasks [refs]. Two-stage pipelines that pair an object detector with a segmentation model have proven particularly effective for nanoparticle analysis [Genc et al.]. However, the practical impact of these advances on routine laboratory workflows has been limited by several factors. Most published methods exist as research code validated on curated datasets rather than as distributable software, requiring users to configure Python environments, manage GPU dependencies, and adapt scripts to their specific imaging conditions. Models trained on particular particle morphologies may fail silently on novel chemistries, overlapping structures, or imaging artifacts common in practice [ref]. And purely automated pipelines typically offer no mechanism for researchers to inspect and correct individual measurements before incorporating them into results; this makes it harder for the quantitative accuracy to be defensible.

Classical image analysis techniques, particularly pixel classification with random forests, occupy a complementary niche. These methods require no pre-trained weights, construct their training signal interactively from the image under analysis, and generalize well to unfamiliar contrast conditions [Weka, LABKIT]. Their principal limitation is in instance-level discrimination: a pixel classifier can separate foreground from background effectively but cannot, on its own, resolve individual particles in a dense or overlapping field. Recent tools such as LABKIT [Arzt et al.] have demonstrated that combining classical pixel classification with modern interface design can produce accessible and practical workflows for biological image analysis. An analogous integration for quantitative materials microscopy, which couples deep learning instance segmentation with classical recovery of missed detections, interactive refinement, and calibrated measurement, has not been realized in an end-user tool.

We present a desktop application for automated nanoparticle analysis from TEM images that addresses this gap. The system combines a two-stage deep learning pipeline, consisting of YOLOv8 for object detection and SAM for pixel-accurate instance segmentation [this is Adra Genc et al exact model/weights], with a random forest pixel classifier that identifies particles missed by the primary detector and converts them into segmentation prompts. All outputs are exposed through an interactive refinement interface in which researchers can correct, split, merge, and re-segment

individual masks before export. Per-particle morphology statistics, including Feret diameter, area, aspect ratio, circularity, and solidity, are computed in calibrated physical units derived from automatic scale bar detection and are recomputed after every edit, ensuring that exported measurements always reflect the current segmentation state. The entire workflow is packaged as a standalone desktop application that requires no programming knowledge, cloud infrastructure, or dedicated GPU hardware and operates entirely offline. In a comparison against manual polygon annotation on a representative micrograph, the platform reproduced the ground-truth size distribution to within 1% of the mean Feret diameter while reducing total analysis time, including interactive refinement, from 48 minutes to under two minutes.

Benchmarking against manual straight-line annotation

To evaluate our platform against an established manual workflow, we compared particle size measurements obtained from four independent sources: manual polygon annotation (ground truth), the Fiji straight-line tool, the platform’s initial automated segmentation, and the platform’s output after interactive refinement.

Dataset

Figure 1 shows the source micrograph used for this comparison. The imaged particles are approximately spherical, with an aspect ratio close to unity for the 30 particles summarized in Table 1 and are largely isolated and homogeneous in appearance. These properties make the field a tractable target for a range of segmentation approaches while still spanning the full size range present in the sample.

Annotation protocols

Ground truth was established by manually annotating every visible particle as a closed polygon, yielding $n = 277$ instances. For the manual baseline, Fiji’s straight-line tool was used to draw a line between the two most distant visible points of each particle; because the particles are approximately spherical, this line length serves as a direct proxy for particle diameter. This was carried out for $n = 50$ particles, reflecting the time cost of manual, line-by-line annotation.

The same micrograph was then processed with our platform. The initial, unedited segmentation yielded $n = 271$ particles. We then applied the platform’s interactive refinement module to correct residual segmentation errors, increasing the count to $n = 275$ particles.

For all four sources we computed a single comparable length metric: the maximum caliper (Feret) diameter, defined as the greatest pairwise distance between vertices of each particle’s convex hull. For the ground truth and platform outputs, this was computed directly from the annotated or segmented polygons; for Fiji, the manually drawn line length was used directly, as it is itself a caliper measurement.

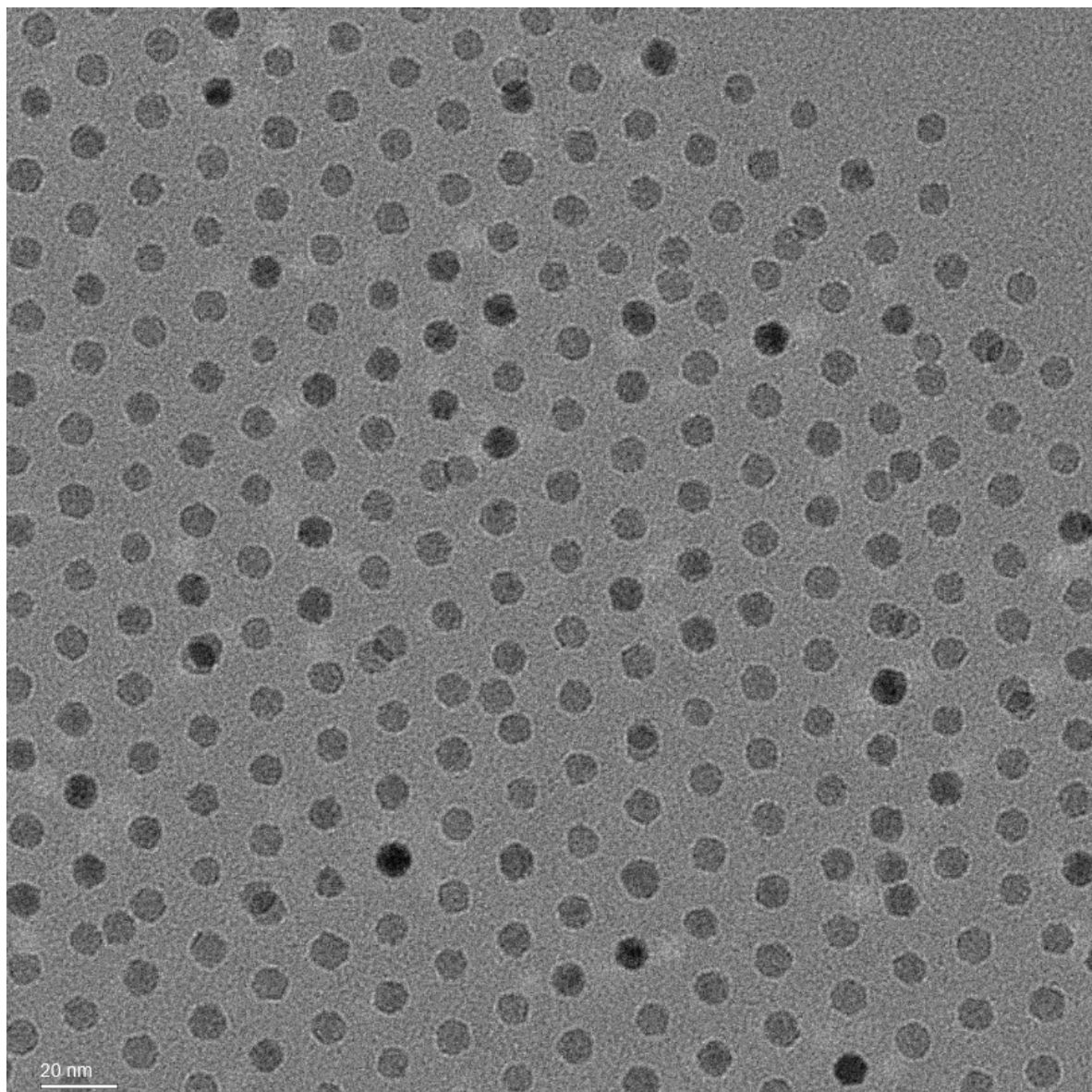


Figure 1: Source micrograph and particle-level statistics. Representative micrograph used for the comparison.

Particle	Longitudinal (px)	Latitudinal (px)	Aspect ratio
1	25.020	23.022	1.087
2	23.087	20.224	1.142
3	22.023	21.024	1.048
4	22.204	25.020	1.127
5	20.000	20.025	1.001
6	23.195	26.019	1.122
7	21.000	26.077	1.242
8	22.000	23.022	1.046
9	21.213	22.136	1.044
10	26.019	25.318	1.028

Table 1: Longitudinal and latitudinal caliper lengths for $n = 10$ ground-truth particles, measured manually in Fiji. Aspect ratio is defined as the ratio of the larger to the smaller of the two measurements per particle (mean = 1.089, maximum deviation from unity = 0.242), consistent with the approximately spherical morphology of the imaged particles.

Results

Figure 2 shows the resulting length distributions for all four methods as a kernel density estimate, an empirical cumulative distribution function, and a box plot with jittered points. The platform’s initial output (mean = 26.98 px, median = 26.57 px, $n = 271$) closely tracks the ground truth distribution (mean = 26.63 px, median = 26.63 px, $n = 277$), and refinement brings the distribution marginally closer still (mean = 26.87 px, median = 26.57 px, $n = 275$). The Fiji manual measurements (mean = 23.93 px, median = 24.58 px, $n = 50$) are shifted toward smaller values and span a narrower range, consistent with a small, manually selected subsample rather than a full-population measurement.

Annotation time differed by more than two orders of magnitude across methods. Manual polygon ground truth required 48 min for 277 particles, the Fiji line-tool baseline required 3 min for 50 particles, and the platform produced its initial segmentation in 6.11 s, with interactive refinement adding a further 97 s ($n = 275$ after refinement, 103.11 s total). Despite this reduction in annotation time, the platform’s size distribution remained closely aligned with the manually annotated ground truth.

The results are summarized in Table 2.

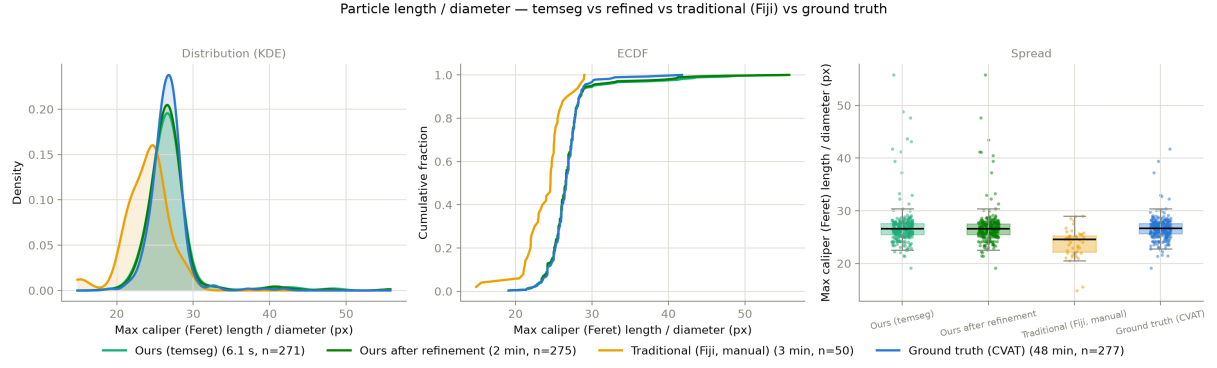


Figure 2: Particle length / diameter distributions across all four annotation methods – ground truth, Fiji manual straight-line, initial platform segmentation, and platform output after refinement – shown as (left) a kernel density estimate, (middle) an empirical cumulative distribution function, and (right) a box plot with jittered points. Legend entries report annotation time and sample size for each method.

Method	n	Annotation time	Mean (px)	Median (px)	Std. dev. (px)
Ground truth (manual polygon)	277	48 min	26.63	26.63	2.21
Fiji (manual straight-line)	50	3 min	23.93	24.58	2.76
Ours – initial segmentation	271	6.11 s	26.98	26.57	3.79
Ours – after refinement	275	103.11 s	26.87	26.57	3.47

Table 2: Summary of particle length / diameter measurements and annotation time across the four methods compared in Fig. 2. Length is the maximum caliper (Feret) diameter for the ground truth and platform outputs, and the manually drawn line length for Fiji. Annotation time is wall-clock time to produce each result set; the platform’s refinement step is reported cumulatively (initial segmentation + refinement).